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Identifying emergent themes of students' transition to university mathematics: a qualitative inquiry with a focus on affective factors

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We present the findings from our inquiry conducted between spring 2019 and spring 2020. It focused on women and students from underrepresented or marginalized populations as they navigated moments in their transition from school to university mathematics, or the secondary-tertiary transition (STT) in mathematics. We draw on Di Martino & Zan's (2010) three-dimensional model of attitude (TMA), in which they reify the role of various affective factors of the STT. We used this model to capture the three affective dimensions of the STT: students' vision of mathematics, their perceived competence in mathematics and their emotional disposition toward mathematics. Our qualitative data included participant interviews, surveys, seminar reflection diaries, seminar recordings and other student artifacts gathered from seminar sessions. In our research, we found differences and commonalities with regard to these three dimensions of the TMA between students who pursued a pure mathematics (PM) degree and those who pursued a combined degree in PM and secondary mathematics teaching (SMT) in the USA. We discuss future directions that can aid women and other historically marginalized students in navigating during STT.

1. Introduction

The secondary-tertiary transition (STT) has been the subject of the study in a variety of contexts outside of the USA (Gueudet, 2008; Clark & Lovric, 2009; Kosiol *et al.,* 2019), and this phenomenon is often associated with mathematics instructors' ongoing perception regarding undergraduate students lacking foundational knowledge or insufficient mathematical preparation on entry at university (Bergsten & Jablonka, 2015). Over time, this knowledge gap has been referred to as the 'transition problem', which includes three related aspects: (a) the conceptual change and learning processes that students experience

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(Dubinsky, 1991; Sfard, 1991; Tall, 1999), (b) the transitions that occur as students move between social groups or contexts with different mathematical practices (Gueudet *et al.,* 2016), such as those experienced when students shift from taking mathematics courses intended for a variety of disciplines at university to those specifically for the study of mathematics and (c) unproductive coordination and dialogue across educational levels that lead to the knowledge gap (Thomas *et al.,* 2015). Thus, the mathematics that students learn and *how* they learn it can change dramatically when they move from secondary school to university, and such differences have the potential to significantly impact students' decision to persevere in the major at university.

The STT in mathematics has been studied from multiple perspectives and in a variety of contexts, primarily by researchers from European countries (see Di Martino *et al.,* 2022). Regardless of this contextual aspect, the difficulties that mathematics majors face are related to cognitive–epistemological, socio-cultural and didactical factors (De Guzman *et al.,* 1998). Recently, there has been renewed interest in affective factors that also play an important role during the STT (Bampili *et al.,* 2017; Di Martino & Gregorio, 2019). Because affective factors are intertwined with the cognitive processes, it is possible such factors can hinder learning if they are neglected (Jaber & Hammer, 2016). Furthermore, in our view, the STT may actually exist as an amalgam of multiple transitions (Alcock & Simpson, 2002), which makes the experiences of individuals unique.

The intersectionality of identities (Crenshaw, 1989) magnifies the complexity of the STT phenomenon and inspires further exploration of the salient elements of students' lived experiences in the STT, particularly women and other historically marginalized students. Gender issues in mathematics, and more broadly in Science, Technology, Engineering, and Mathematics (STEM), have long been debated. Very few studies address gender from the standpoint of STT research, particularly in mathematics (e.g. Rodd & Bartholomew, 2006). Due to the predominance of masculine discourse in numerous mathematical spaces, women are often recognized as marginalized in such mathematical communities (e.g. Mendick, 2005). Solomon (2007) reported that women in mathematics voiced identities of exclusion and did not perceive themselves as negotiators of meaning in discursive spaces. Furthermore, women felt that they were required to do 'identity work' to feel they could participate as legitimate actors in mathematical communities at university, particularly given the long-established majority male presence in the field of mathematics (Solomon, 2012, p. 180). As such, in our study, we used the term 'marginalized', encompassing all women, with some confronting a double disadvantage due to their racial or ethnic background. Men of various racial and/or ethnic backgrounds—who can also be considered as underrepresented in mathematics (e.g. in the context of the US, African American and Hispanic men)—were also included as marginalized within the population of our study.

Our primary goal for this research was to explore the features of the STT phenomenon in mathematics within an institution of higher learning in the United States. We emphasize the importance of the context in our study due to the paucity of such examples in the literature that focus on this phenomenon. In this article, we describe a pilot study we conducted with the goal of understanding the ways in which women and other historically marginalized undergraduate mathematics students reflect on affective aspects of perceived competence and vision of mathematics (Di Martino & Zan, 2010) while they navigate the secondary-tertiary transition. Moreover, we discuss the tensions in students' perceived competence and vision of mathematics by highlighting the emotional dimensions that arise as part of experiencing this phenomenon. To accomplish this, we also made a distinction between secondary school mathematics teaching (SMT) majors and pure mathematics (PM) majors to account for differences in their STT experiences. Therefore, the following overarching research question guided our inquiry: *How do women and other marginalized students who are either PM or SMT majors experience university mathematics with regard to affective aspects of the STT?*

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2. Background and theoretical perspectives

To understand the STT in mathematics from the student perspective, we draw on ideas from the affective domain, conceptualized through beliefs, attitudes and emotions by McLeod (1992). The investigation among and between these three constructs constitutes research focused on affect (Di Martino & Zan, 2011). Through their investigation of the affective factors, Di Martino & Zan (2010) expanded on McLeod's ideas and conceptualized mathematical attitudes as a bridge between mathematical beliefs and emotions. In this work, we adapt Di Martino & Zan's (2010) three-dimensional model of attitude (TMA), which reifies the role of various affective factors in the STT with a focus on students' emotional disposition toward mathematics, perceived competence in mathematics and vision of mathematics. Our research goal was to investigate students' experiences during the STT in mathematics, focusing on how they deal with certain affective factors such as their beliefs about themselves as learners (i.e. perceived competence), their beliefs about the nature of mathematical knowledge (i.e. vision of mathematics) and emotional disposition toward mathematics.

Affective factors, according to Mandler (1989), are the result of emotional reactions to a disruption in a routine (McLeod, 1992). Similarly, the STT in mathematics is also considered as an interruption in a routine (Clark & Lovric, 2009) and/or represents discontinuities (Gueudet, 2008) between the two institutions. When students experience a 'life crisis' in 'a rite of passage', such as transitioning to university mathematics, their typical routines are 'interrupted, changed and distorted' (Clark & Lovric, 2008, p. 26). For example, when students move to university mathematics, they are introduced to new routines in modes of thinking and knowledge organization for the construction of proofs (Gueudet, 2008). Accordingly, various affective factors arise, such as feelings of inadequacy and frustration, as a result of 'the cognitive evaluation of interruption' in mathematical learning situations (McLeod, 1992, p. 578). Thus, cognition and emotions interact deeply (Di Martino & Zan, 2011). Briefly, the similarities between the emergence of affective factors in a life crisis and students' affective responses to challenges during the STT underpin our current research into affective factors in the STT.

Di Martino & Zan (2010) examined students' emotional dispositions toward mathematics in relation to their perceived competence and vision of mathematics. That is, when students experienced difficulties in mathematics, they often expressed those difficulties with certain emotions ranging from positive to negative, such as enjoyment, interest, pride, frustration, shame, intimidation, inadequacy and so forth (Di Martino & Gregorio, 2019). Furthermore, according to Op't Eynde *et al.* (2006), recognizing unproductive[1] emotions in the situated context can aid researchers in developing an in-depth understanding of underlying factors. Di Martino & Gregorio (2019) identified students' beliefs about mathematics and beliefs about themselves as learners by investigating their emotional responses. Students' reflections on their beliefs specific to the nature of mathematics (vision of mathematics) and their mathematical competency (perceived competence) provided insights into their emotional disposition toward mathematics. As a result of the interrelationship of the three components (i.e. vision of mathematics, perceived competence and emotional disposition toward mathematics), Di Martino & Gregorio (2019) were able to conceptualize affective experiences during the STT in mathematics.

According to McLeod (1992), students' beliefs about themselves as mathematics learners are one of the major aspects of their affective experience. Research around beliefs about self involves investigation

[1] Although some may conceptualize emotions such as a dislike of mathematics as 'negative', we choose to consider these as 'unproductive' emotions. That is, we consider such emotions (e.g. dislike, frustration, confusion) as inhibiting or adversely impacting a student's disposition related to mathematics and thus causing unproductivity with or in mathematics—even if temporarily.

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of self-concept (e.g. Rach & Heinze, 2017), perception of competence and confidence, considering differences in students' gender, race and/or ethnicity. In our research, we explore students' perceptions of themselves in mathematics learning situations using the perceived competence dimension of the TMA. Di Martino & Zan (2010) examined perceived competence in mathematics from the student perspective and associated the concept with one's attribution of success or failure (i.e. idea of success, as articulated by Di Martino and Zan) to obtaining good school grades and a level of understanding of mathematics (i.e. meaning of understanding, as expressed by students). For example, university students attributed their 'failure' in examinations to internal factors such as their perception of mathematical competency rather than differences in the nature of mathematics at university compared with secondary school (Furinghetti *et al.,* 2013; Di Martino & Gregorio, 2019). The consideration of students' conception of success (or failure) in their study of mathematics helps to exemplify that across the three dimensions of Di Martino & Zan's (2010) TMA, it is quite difficult to treat any one in isolation. For example, students' perceptions of their competence in mathematics may be layered in different emotional dimensions that can be linked to other aspects of their learning of mathematics, such as expression of understanding (or levels of understanding). Importantly, as Di Martino & Zan (2010) observed, 'the link between emotional disposition and perceived competence in mathematics is very complex *because* the perceived competence is linked to one's idea of success in mathematics' (p. 40, emphasis added).

Di Martino & Zan (2010) found that students referred to at least 'one or more of these dimensions' (p. 39) when they talked about their stories with mathematics. They claimed that the idea of success can change for students, which makes the analysis of associations between emotions and perceived competence more complex. Even after distinguishing the idea of success, whether it is about getting good grades or understanding of mathematics, the meaning of understanding also changes depending on the students' vision of mathematics. Thus, making sense of students' experiences in-depth around the three dimensions is critical for a nuanced analysis.

Di Martino & Zan (2010) aligned their conception of vision of mathematics with Skemp's (1976) conceptualization of relational (problem-solving process and relations in mathematics) and instrumental (rules without reasoning) mathematics. Di Martino & Zan (2010) documented that a negative disposition toward mathematics was often linked to an instrumental vision of mathematics or low perceived competence. The connections between the vision of mathematics and emotional disposition also implicitly relate to students' perceived competence in mathematics. As a result, rather than analysing the dimensions independently, the authors suggested examining students' mathematical experiences by addressing the interconnection among them. For our research, we broaden Di Martino and Zan's original conception of vision of mathematics as a conceptualization of the four views on the nature of mathematics described by Grigutsch & Törner (1998). These views include schema (mathematics as a collection of procedures and rules), formalist (mathematics characterized by rigor, formal logic and the ability to abstract and formalize), application (mathematics provides capacity to solve problems of practical use in society) and process (mathematics relational thinking as problem solving) aspects. While the first two beliefs are aligned with static viewpoints, the latter two represent dynamic viewpoints in mathematics (Grigutsch & Törner, 1998). It is worth noting that mathematical beliefs are not restricted to these characteristics; they can manifest as a combination of two or more of these aspects. To further support our choice of conceptualizing vision of mathematics, more broadly as views on the nature of mathematics, we believe Skemp's instrumental thinking is aligned with orientations similar to Grigutsch and Törner's schema aspect and relational thinking as reminiscent of more dynamic aspects, such as application and process. Therefore, we consider students' views on the nature of mathematics as being synonymous with their vision of mathematics.

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TABLE 1. *Transition research participants (Spring 2019–spring 2020)*

Student	Year at university†	Major	Race/Gender	Participation
Bethany	1	SMT	White/female	Interview only
Dana	1	PM	Hispanic/Latina/female	Interview; 2020 seminar
Manuel	1	SMT	Hispanic/male	Interview; 2020 seminar
Amelia	2	PM	Hispanic/Latina/female	Seminars (2019, 2020)
Karine	2	PM	Hispanic/Latina/female	Interview; 2019 seminar
Sunny	2	PM	White/female	Interview; 2020 seminar
Diedre	3	SMT	White/female	Interview only
Malya	3	SMT	Black/female	Interview; 2019 seminar
Amber	4	PM	Hispanic/female	Interview; 2019 seminar
Cynthia	4	PM	White/female	2019 Seminar
Evelyn	4	SMT	White/female	Interview; 2019 seminar
Tanya	4	SMT	White/female	2019 Seminar
Tina	4	PM	Black/Female	Interview only

†The reported 'year at university' (1, 2, 3 or 4) refers to a typical classification in the USA: freshmen (1), sophomore (2), junior (3) and senior (4), and for this study, corresponds to a student's year at university for their initial participation.

To further declare our position regarding students' vision of mathematics, we subscribe to the idea that, whereas the nature of mathematics tends to be empirically oriented (i.e. object-schema) in secondary school, it is more formal and abstract at university (Witzke *et al.,* 2018), particularly in upper level mathematics courses. This change in the vision of mathematics (i.e. beliefs on the nature of mathematics) constitutes another conception of the transition for undergraduate mathematics majors. When this shift in the nature of mathematics appears 'suddenly', students recognize a transition in sometimes significant, emotionally laden ways, which is manifested through reflecting on emotional dispositions (Witzke *et al.,* 2018). Therefore, we presumed that if the disruption in the routine is especially uncomfortable, we would also detect emotional responses (e.g. frustration, apprehension and dislike) in our analysis of students' experiences in the STT in mathematics.

It is important to note that Di Martino & Zan (2011) argued that attitudes develop through the interplay between beliefs and emotions in studying mathematics and offered a useful lens to examine affective experiences during the STT. The three dimensions of the model and the interplay among and between them have deepened our understanding of students' experiences with mathematics during the STT regarding the role of affective factors. As a result, we believe that the three dimensions of the TMA can be useful to investigate the STT phenomenon.

3. Method

3.1. Context and participants

In total, 13 mathematics majors who were enrolled in a large research university in the southeastern United States participated in the study in 2019–2020 (see Table 1). Our data sources included participant interviews, recordings of seminar sessions, surveys, seminar reflection diaries and other student artifacts collected from seminar sessions. In our investigation, we studied mathematics majors who were at different points during the STT, with a focus on affective factors in relation to their beliefs, attitudes and emotions. More specifically, interview protocols and seminar content explicitly addressed the dimensions of the TMA: vision of mathematics (i.e. beliefs on the nature of mathematics), perceived competence in mathematics (i.e. beliefs about abilities) and emotional disposition toward mathematics.

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The goal of our research (and our primary research question) precipitated our purposeful participant selection. In spring 2019, there were 330 mathematics majors[2] at the university where we conducted this study. Of these (and with permission granted by our institutional review board), 96 students identified as majoring in either pure mathematics (PM) or double majoring in pure mathematics and secondary mathematics teaching (SMT). We purposefully selected students from these two majors because we were particularly interested in students who would potentially have more frequent experiences with proof courses in the major, and at the institution in which we conducted the study, these were PM and SMT majors. Of these 96, 66 were women and/or identified as non-White. Finally, after contacting all 66 in our population of interest, we obtained a purposeful convenience sample of 13 students (12 women, 1 man), nine of whom volunteered to participate in a 1-h interview (Table 1).

3.2. Data sources and data collection

Interviews were designed to capture students' experiences regarding the STT using the dimensions of the TMA and surveys available in the relevant literature. The interview protocol included questions regarding student information, secondary school mathematics experiences (Smith, 2012; Witzke *et al.,* 2016; Witzke *et al.,* 2018), decision to major in mathematics (Smith, 2012), current university mathematics experiences (Smith, 2012; Thomas *et al.,* 2015; Witzke *et al.,* 2018), perception of mathematics (Grigutsch & Törner, 1998; Dubinsky & McDonald, 2001), perceived confidence in mathematics (Liston & O'Donoghue, 2010) and approaches to and assessment of learning mathematics (Liston & O'Donoghue, 2010; Thomas *et al.,* 2015; Di Martino & Gregorio, 2019) (see Appendix for sample questions). Interviews were critical for understanding students' beliefs about the nature of mathematics in relation to their perceived competence, and to prompt them to reflect on their emotions relevant to the STT.

Next, in 2019, we designed and implemented a four-part (each for 3 h in duration) seminar, 'Women and Underrepresented Minorities in Mathematics', in which seven students participated. Before the start of the seminar, students completed an online survey, with demographic questions and questions about their perceptions of the nature of mathematics, mathematics content-related items and The Mathematics Attitudes and Perceptions Survey (MAPS) items (Code *et al.,* 2016) to examine students' attitudes toward mathematics. MAPS items were a 5-point Likert scale format and measured students' expert mathematics views related to their perceptions and attitudes about mathematics. The 2019 seminar included discussions, activities and reflections for the purpose of capturing the ways that students talk, think and feel about mathematics as a discipline. We also prompted students to discuss their views on the nature of mathematics by emphasizing those of mathematicians throughout the history of mathematics. In addition to collecting student survey responses and video-recording seminar sessions of discussions, students were asked to complete written reflection diaries (collected via Google documents) each day to reflect on their beliefs, attitudes and emotions, and the reflection prompts corresponded to the content of the different seminar sessions. The 2019 seminar was instrumental in our exploratory research designed to make sense of the influence of different aspects of the STT for students completing a mathematics major at an institution in the USA.

[2] At this university, students can declare one of five mathematics majors: actuarial science, applied/computational mathematics, biomathematics, pure mathematics and pure mathematics and secondary mathematics teaching (as a double major).

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The 2020 seminar focused on students' perceptions of the characteristics of mathematicians, with an emphasis on mathematical identity. We also addressed students' perception of available institutional support. Four students participated in the 2020 seminar[3] comprising four weekly sessions (each for 2 h in duration) and for which data were collected in the form of shared Google documents from session activities and video-recorded seminar sessions. We hoped to capture students' views on why the nature of mathematics matters (in the study or practice of mathematics), as well as their conceptions regarding attributes of mathematicians. In this way, we were especially cognizant of the question raised by Geisler & Rolka (2021) regarding 'which beliefs actually fit to university mathematics and therefore are beneficial during the transition' (p. 603). Furthermore, we also intended to provide students with a form of institutional support by creating a small mathematical community as part of the seminar. We adhered to the three dimensions of the TMA (Di Martino & Zan, 2010)—accounting for our modification on students' views on the nature of mathematics—to reflect on students' perceived competence and emotional disposition toward mathematics, along with institutional aspects on the STT.

Finally, students were asked to complete an online survey at the conclusion of the 2020 seminar. The survey was similar to the survey implemented during the previous seminar in that it captured demographics and students' conceptions of an expert view on mathematics as a discipline. As well, we asked students to identify similarities and differences between the nature of mathematics at secondary school and university and for their perceptions of fundamental concepts in university mathematics (e.g. definitions, axioms, theory, proofs). The second survey also addressed themes from our discussions during the 2020 seminar, including prompts on students' use of supportive programs or resources (academic and emotional) and attributes of mathematicians.

3.3. Data analysis

We transcribed the interviews using the f4transkript software. Adopting aspects of the grounded theory methodology (Charmaz, 2006), we analysed our data using open, axial and selective coding. The first author performed data analysis for interviews, while the second author conducted data analysis for reflection diaries and seminar discussions. Using the components of our theoretical framework, we sought to identify the themes and patterns that emerged from the responses of the participants. As researchers, we sought patterns that '(a) added evidence of the same pattern, and (b) remain open to disconfirming evidence when it appeared' (Miles *et al.,* 2020, p. 275). For the open coding stage, we noted numerous quotes from participants that addressed STT broadly including cognitive, didactic, sociocultural and affective aspects. For example, we documented multiple difficulties students noted related to change in the environment and nature of mathematics at university when compared with high school mathematical experiences. During axial coding, we narrowed our focus to the three dimensions of the theoretical framework to capture links among and between them. For example, we focused on the underlying factors that led to a decrease in students' perceived competence and we looked for emotional responses accompanied in their reflections. In the selective coding stage, we paid attention to comparing and contrasting patterns in students' experiences. For example, we noted differences between the perception of SMTs and PMs in our data analysis. In addition, we also recorded pertinent participant quotations for use in reporting our findings. The authors met on a weekly basis to share

[3] This seminar began just before when the global COVID-19 pandemic first impacted the USA and continued until the end of the spring 2020 semester. Consequently, the first two sessions of the seminar were held on campus, but the third and fourth sessions were held via Zoom.

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insights gleaned from independent data analysis procedures and to mitigate potential researcher biases. During the iterative data analysis processes, we also referred to our researcher memos and field notes as resources to support emergent themes.

4. Findings and discussion relevant to dimensions of the TMA

We set out to examine affective aspects during the STT in mathematics along two foci, and thus we arranged our findings as follows. In the first, we consider two pairs of aspects directly from the TMA (Di Martino & Zan, 2010). We begin with a presentation of findings and associated discussion for the aspects of vision of mathematics and associated emotional dimensions. Then, we do the same for the aspects of perceived competence and emotional dimensions. For the second focus, we identify themes that emerged from the data that alerted us to students' experiences with relevant support mechanisms, both social–emotional and academic, and either those actually experienced or desired, and for which we are particularly interested from the perspective of emotional dimensions, to maintain our connection to the TMA.

4.1. Vision of mathematics and emotional dimensions

All majors noted similarities between the application aspect (Grigutsch & Törner, 1998) of university mathematics and secondary school mathematics. Interestingly, students highlighted similarities in application aspects more often, such as: 'The similarities [between school versus university mathematics] are more apparent in the applied mathematics courses I have taken rather than the pure ones at the university-level' (Dana, Survey 2020). As Dana's reflection illustrated, the lower division[4] mathematics courses were an extension of high school mathematics courses. PM majors and SMT majors recognized a shift in the nature of mathematics between secondary school and university mathematics. That is, whereas students might describe the nature of mathematics in secondary school as being driven by applying a collection of rules to solve exercises, university mathematics is perceived as demanding deeper thinking:

> . . . the real transition didn't really happen for me until I started to take...math classes [in the major]. Because I feel for the most part the [Calculus sequence was] pretty much formula-based and the things were still tangible. . . . we're still using numbers, whereas now I feel it's a completely different strain of math; it is all theoretical; it is all proof-based and it is just something very different. (Evelyn, Interview, April 2019).

This quote from Evelyn's interview indicated that previous mathematics knowledge in secondary school may not be a predictor of success in university due to a disconnect between two scientific knowledge (Rach & Heinze, 2017), particularly in upper division courses. Evelyn's and Dana's responses also demonstrate a distinction when comparing the STT experiences of students in the USA to those of students in other countries.[5] Students noted a shift in when 'the real transition' occurs; that is, not immediately on taking a first mathematics course or courses at university, but at some point later, when taking courses in the mathematics major.

[4] Lower division courses are those designated as typically taken in the first or second year during an undergraduate program; upper division courses are typically taken during the third and fourth year.

[5] For example, in a case study conducted in Germany, Inga experienced 'initial, sudden difficulties in mathematics' (Witzke *et al.,* 2018, p. 71) on transitioning from secondary school to a first course at university (in her case, Analysis).

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We identified some distinctions between SMT and PM majors' emotional dispositions toward the abstractness of mathematics. Whereas SMTs perceived this new abstract phase of mathematics as a burden, PM majors elaborated on this as a source of enjoyment. For example, Amelia reflected that "the proofs I was introduced to [in secondary school] were the trigonometric proofs, and up until that point, they had been the most fun thing I had done in mathematics" (Reflection Diary, June 2019). Thus, Amelia demonstrated enjoyment and excitement when she was talking about the nature of mathematical knowledge at the university level. Similarly, Sunny expressed that "[I enjoy] proofs because it feels like putting together a puzzle that you know you have the pieces for. It's an exercise in logic and creativity" (Survey, April 2020). These findings supported Kosiol *et al.*'s (2019) claims that students with an interest in proofs and formal mathematics tend to enjoy university mathematics more than those with an interest in school mathematics.

Students reflected on their emotions as a result of the changing nature of mathematics in university. Similar to Di Martino & Gregorio (2019), the students in our study, who are 'successful' in the major, also reflected conflicting emotions ranging from positive to negative. Although Dana expressed similar sentiments to Amelia and Sunny about how much she enjoyed studying mathematics at university, she elaborated on the broad nature of mathematics as being sometimes the least enjoyable characteristic of mathematics:

> I may not enjoy every second of studying mathematics, some concepts may not be clicking for me or there may be an idea that seems impossible to visualize, yet mathematics continues to challenge me as I persist and imagine this self-contained world that is so infinite in its nature, and that no matter how much I learn from it, there will always be more to learn—and hopefully one day to discover. Mathematics challenges me to think in ways that I have never done before, to push myself to try and imagine the unimaginable, which is what I love/enjoy about it so much . . . This one is hard to say, but I think the idea of mathematics being so abstract in its nature . . . is also, in a way, what I enjoy the least about it. (Dana, Survey, April 2020).

In this quote, Dana's 'least enjoyment' toward studying mathematics related to her vision of mathematics was not necessarily counterproductive to her success in mathematics. Furthermore, in this case, we use the characterization of 'least enjoyment' from Dana's words, 'I may not enjoy every second of studying mathematics . . . ' However, throughout our data collection and analysis, we came to understand this expression as less of a negative disposition toward the whole of the mathematics discipline. Instead, we considered this as a case of unproductive emotions, in that Dana held a view about the challenging nature of mathematics, but it was not one which caused her to completely shut down in terms of her pursuit of or persistence in mathematics. Rather, Dana's interest and desire to comprehend mathematics pushed her to persevere even in the face of adversity in learning abstract mathematics. Part of the struggles Dana experienced during her transition seemed to originate in difficulties with proofs pertaining to the changing nature of mathematics (Furinghetti *et al.,* 2013; Thomas *et al.,* 2015). This new (i.e. abstract) phase of mathematics at university revealed the affective aspects of the STT.

From another perspective, Evelyn discussed the abstract nature of mathematics in such a way that seemed to hinder her learning of advanced mathematics:

> I feel like I don't really understand most of it [upper division mathematics courses]. Because I feel like a lot of it is very abstract. And I feel like a lot of professors don't elaborate [...] on some things. So I learn the very basic [material] but then, I don't know anything else about it or where, when to use it even . . . Yes, I can employ some of the strategies that I am learning in them. But for the most part, they are [abstract concepts] not gonna benefit me really in teaching that kind of stuff. (Evelyn, Interview, April 2019).

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Evelyn's disposition toward advanced mathematics was relatively on the unproductive side, which made her feel confused about how to use it. In addition, Evelyn critiqued professors' approach to the teaching of advanced mathematics. Thus, her frustrations seemed to be coupled with pedagogical and cognitive factors (Thomas *et al*., 2012). Evelyn's vision of mathematics seemed to also be aligned with the belief that Schoenfeld (1985) identified: 'Formal mathematics has little or nothing to do with real thinking or problem solving' (p. 43). That is, Evelyn merely associated advanced university mathematics with the 'school subject' (Di Martino & Zan, 2011, p. 477) and questioned the practical use of abstract mathematics in society. In other words, she tended to view mathematics from an application-oriented perspective, which alone did not support her success in upper division mathematics classes (Geisler & Rolka, 2021). Evelyn's views could also hint at her vision of mathematics regarding secondary school mathematics. Evelyn perceived school mathematics and university mathematics as two related yet distinct subjects, and she may have experienced difficult moments during the STT in mathematics (Rach & Heinze, 2017). Thus, Evelyn opted for the belief that formal and abstract mathematics were not useful for her future teaching practices because she perceived certain discrepancies between the nature of mathematics at secondary school and at the tertiary level.

On the other hand, although having certain emotional dispositions, PMs reflected on the abstract nature of mathematics at university with a noticeable level of excitement and curiosity. We did not notice the recurrent questioning of the real-life relevancy of abstract mathematics with PMs, which we observed with SMTs. The difference in their beliefs about abstract mathematics between SMTs and PMs may also be due to the fewer number of advanced mathematics courses taken by SMTs. Evelyn's continuous frustration in advanced mathematics courses—given her perception of their lack of utility in her future teaching—may have led her to believe that advanced mathematics is not accessible for her, directing her to alter her personal goal in this area (Hannula, 2002, 2006). SMTs' experiences raised concerns regarding the relevance of advanced mathematics courses to the practices of prospective mathematics teachers (Even, 2011).

We assert that students' future aspirations for mathematics also influences their idea of success. The differences between PMs' and SMTs' vision of mathematics and pertinent emotional dispositions toward learning abstract mathematics could be due to the students' meaning-making of their mathematics associated with their future career goals. That is, SMTs were concerned about translating abstract ideas into teaching situations to improve their pedagogical practices (Wasserman *et al.,* 2019). PMs seemed to focus primarily on attempting to grasp abstract concepts to develop the characteristics of prominent mathematicians while placing a strong emphasis on their own learning. In this way, we find the PM students positioned as apprentices 'whose training begins with an exposure to the master's [mathematician's] discursive practices' (Sfard, 2014, p. 201), with their learning dependent on 'trying to figure out what it is that the master would have done in a similar situation' (Sfard, 2014, p. 202).

4.2. Perceived competence and emotional dimensions

SMTs and PMs discussed their perceived competence in mathematics differently, although there were overlaps in reflections about their confidence in mathematics. It was remarkable that all participants had pleasant mathematics experiences in secondary school and thus developed high confidence in the subject. School grades emerged as an important and meaningful factor on students' mathematical confidence (Liston & O'Donoghue, 2010; Pampaka *et al.,* 2011). SMTs tended to rely on grades as a measure of competence in mathematics, which also seemed to have detrimental influences on their confidence in

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mathematics, especially in university mathematics. PMs reported that receiving high grades boosted their confidence, yet grades alone were not strong enough criteria to determine the level of competence in mathematics. Sunny, who received grades of more than 90% (of 100%), demonstrated conflicting patterns in her beliefs about confidence. That is, Sunny was not certain whether she considered herself as confident even though she earned high grades.

> Sunny: I never feel competent enough if that makes sense. I know that I am competent, but I am always . . . I don't, yeah . . . Like, I know that if I decided to achieve a PhD, probably I could. But I also don't feel like I could.
>
> Interviewer: Why?
>
> Sunny: Umm. I don't know. I just feel like you just have to be really good at math. I know from . . . just from grade aspects of it, I am really good at math, but I never feel like I am *that* good at math. Like good enough to have a PhD if that makes sense [...] I guess . . . I don't feel like (sighs deeply) my grades say, 'hey you are really, really good at math', but my mentality towards it is, I am never gonna be as good at math as these people with PhDs who do this research and then write these books and then understand this. (Interview, March 2020)

This excerpt raised a question regarding the extent to which high grades are commensurable with perceived competence and, thus, confidence in mathematics. Interestingly, although getting a lower grade can be an explicit indication of low competence and even failure from students' perspectives, receiving the highest grade did not seem to be an indication of higher competence particularly for PMs. Pampaka *et al.* (2011) associated low perceived competence in proof comprehension in university with lower previous mathematical attainment, which did not explain Sunny's case of lower perceived mathematical competence. Pampaka *et al.* (2011) also found gender differences favoring male students with stronger self-efficacy beliefs, which disappeared for PM students. However, in this study, Dana and Sunny, who were both PMs, still reported lower perceived competence in mathematics. Sunny reflected conflicting patterns in her perceived competence, which opened up the 'potential for new meaning' (p. 182) for her self-positioning within the mathematics community (Solomon, 2007). When we asked whether Sunny considered herself competent, she started to question whether the mathematics program was easy or whether she was an intelligent student who was capable of earning excellent grades. Our initial findings prompted more questions than answers about why some women, such as Sunny, do not feel good about their competence in mathematics and consequently seek 'counter-claims' (Solomon, 2007, p. 178) that have the effect of lowering their mathematical confidence. These perspectives highlight such conflicting beliefs on perceived competence in mathematics, which do not seem to be merely explained by earning good grades.

The way Sunny articulated her perception of mathematical competence illustrated a negotiation on the *idea of success*. Di Martino & Zan (2010) linked this approach to the relational understanding of mathematics, which goes beyond receiving good grades. Sunny expected herself to be able to understand famous mathematicians' writings to feel competent. This shows that Sunny perceived mathematicians to be 'extremely intelligent', a trait with which she did not identify despite her high achievement in both school and university mathematics (Piatek-Jimenez, 2008, p. 642). These beliefs are consistent with Piatek-Jimenez's (2008) assertion that women hold stereotypical beliefs about being mathematicians, which might prevent them from pursuing advanced degrees in mathematics because they do not identify with particular characteristics they believe competent mathematicians possess. Dana also reflected on her perceived competence, similar to Sunny, by adding peer comparison into the conversation:

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> You put yourself in a room with all the other undergraduate mathematicians, and all of a sudden, like, your confidence just goes all the way down. Because, you know, more than half of them probably know a lot more than you do. And it's, I mean, it's not that that makes you not as confident, but it's almost as if there's so much more that I don't know that all these other people do. And then it's like, I don't know anything. And then you just start second guessing everything you've ever learned. (Dana, Interview, April 2019).

Dana articulated her perceived competence in a way that reflected the big-fish–little-pond effect as a potential aspect of transitioning to university when she was among other high-achieving mathematics students (see Uysal & Clark, 2021a). Di Martino & Gregorio (2019) also documented a similar pattern in their examination of STT in mathematics related to the peer comparison aspect, which further contributed to students' doubts about their mathematical competence.

In contrast to this, Manuel did not resonate with Dana's experiences and stated that he 'found more like a drive and competitiveness about [the idea that they know more than I do] to learn more, basically' (Seminar, April 2020). We noticed differences in Manuel's approach to the idea of success as he described his experiences with mathematics examinations as 'even if [your answers are] right, you feel like, in some sense, I didn't learn it, I just memorized it, or I just got lucky' (Interview, April 2019). When Manuel was prompted about his competence in mathematics, he stated that 'I think [my confidence] is like, very good. I mean, it is not the highest in my opinion. And that's something I always learned, that there's always someone way better than you thought you were' (Interview, April 2019). In addition, Manuel mentioned that he feels good about his competence if he earns a good grade. He did not seem to consider himself held back by not having an understanding of certain mathematical ideas. Manuel's meaning of success appeared different from that of Dana or Sunny. Even though Manuel experienced a decrease in his mathematical competence during the STT in mathematics, he did not attribute it to his overall mathematical confidence. However, both women illustrated conflicting patterns regarding their perceived competence and, consequently, their developing identity as mathematicians. This circumstance is reminiscent of the socially constructed influences regarding gender identity and underrating of mathematical competence (Mendick, 2006). Our case is akin to Solomon (2007) in which tensions, such as those experienced by women believing they have 'less right to be there' (p. 80), could lead to more stress and more questioning of abilities. The social context plays an important role in constructing mathematician identity in relation to societal beliefs, expectations and roles (Piatek-Jimenez, 2008) as well as social and academic demands of the institutions (Hernandez-Martinez *et al.,* 2011). Researchers should take the intersectionality of identities into account when analysing the STT experiences. Given the highlighted circumstance, women and other marginalized groups need additional support mechanisms (e.g. institutional support) during the STT in mathematics for academic and social identity development (Hernandez-Martinez *et al.,* 2011; Mishra, 2020).

4.3. Forms of support and emotional dimensions

As part of our inquiry into students' experience with the STT, we sought to capture which mechanisms they identified as providing social or academic forms of support (Clark & Lovric, 2009). First, students from both majors similarly discussed that their parents and secondary teachers were supportive in their decision to major in mathematics. Second, they mentioned that their high school mathematics background, such as completing advanced placement (e.g. Evelyn, Manuel, Dana, Sunny) and dual enrollment (e.g. Karine) courses assisted with the transition to advanced mathematics courses at university, especially with the computational aspects of mathematics. Amelia stated that she interacted

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with proofs in high school mathematics courses which, she believed, smoothed her transition to abstract courses at university. Amelia's noticeable positive engagement in proof could be attributed to her early exposure and interest in proof (Furinghetti *et al.,* 2013; Kosiol *et al.,* 2019).

Students also identified a need for institutional support when dealing with the abstract nature of mathematics. Students' descriptions focused on accessing external online resources for content support (e.g. Google, YouTube and various software) rather than attending office hours or using university-provided tutoring services. Malya indicated that university-provided tutoring services did not employ tutors with proficiency in upper division mathematics, which she indicated she needed to be successful. However, she shared that a formalized community within the institution specifically designed to provide a transition support program for students of color provided her with other forms of academic and emotional support for a sense of belonging (Mishra, 2020). In addition, Karine and Dana described their membership within student scientific communities (specifically for women in STEM) as a significant support mechanism. Karine, Dana and Evelyn expressed the need for more women faculty and advisors as role models for them to feel supported in and belonging to their mathematics program (Piatek-Jimenez, 2015).

Interestingly, we observed that PM majors were comparatively more inclined to study individually to make sense of complex concepts, whereas SMT majors highlighted their participation in peer working groups to complete assignments. Furthermore, SMT majors strongly and consistently advocated for the importance of necessary improvements in instructional practices and instructor quality. PM majors focused more on internal learning aspects, such as claiming responsibility for most of the difficulties they encountered rather than questioning instructional practices of mathematics professors. For example, at the end of the spring 2019 seminar, Evelyn began to consider abstract mathematics somewhat differently; however, her reflection remained strongly focused on—in this example—her number theory professor's instructional practice:

> After the seminar, I am a little more open to dealing with formal/abstract mathematics. It makes me wish that mathematics in the upper level was taught in a different way. I feel like if it was, I would maybe be able to comprehend more abstract concepts a bit more. For example, in number theory, our professor did little explaining on proofs she would show us, which made it difficult to follow what was happening. I think if there was a little more discussion, explanation or collaboration, like the seminar, then I think I could have gotten more out of the class. (Evelyn, day 4 Reflection, Seminar, June 2019).

In her reflections, Evelyn emphasized how the seminar helped her to recognize that collaboration with peers and instructors could positively contribute to her mathematics experience (i.e. regarding traditional course experiences). She also addressed the need for changing teaching approaches to impact her understanding, particularly in advanced mathematics courses. Amelia, a PM student, also discussed a necessary change in the way mathematics is taught at university:

> I always look at the proofs for the rules and try to understand them, so that I can tell myself that I am not just coldly memorizing something with no understanding. On that note, a lot of people think that mathematics is about memorization, that is, memorization of rules, equations, properties, etc. I think of mathematics as something that has to be understood and not memorized, which is why I get really frustrated when I do have to memorize something because of say, lack of time. (Amelia, day 2 Reflection, Seminar, June 2019).

Amelia's questioning of instructional practices shared a common premise with Evelyn's. Amelia's frustration stemmed from an emphasis on rote memorization rather than understanding. Amelia did not describe support in the form of alternative or improved instructional practices as explicitly as

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Evelyn. Indeed, her description was more internally directed; i.e. she became frustrated when she had to memorize something due to lack of time, and this implies that in some way the lack of time was her fault. Regardless, students expressed heightened emotions when discussing important aspects of their mathematics experience— with regard to instruction and the meaning-making left for them to do as a result of it—and the relevant potential need for supportive mechanisms to assist them in doing so. These STT aspects seemed to be related to the 'didactic contract' between instructors and students, which necessitates an orientation for different ways of thinking to progress to advanced-level mathematical thinking (Pepin, 2014, p. 653).

As well, the seminar also provided a venue that supported students' development, both mathematically and socially (see Uysal & Clark, 2021b). For example, students were afforded the opportunity to weigh in on fundamental questions such as, 'What *is* mathematics?' and 'What are mathematical objects?'—questions that promoted significant discussion, but which students also noted they had not previously had the opportunity to discuss with peers.

5. Concluding remarks and implications for future research

By investigating the link among vision of mathematics, perceived competence and emotional disposition toward mathematics, we sought to understand how undergraduate mathematics majors experienced the affective aspects of the STT. Our study focused on a small population of students pursuing either a PM degree or a combined degree in PM and SMT offered insight into the features of the transition to university mathematics experienced by them. In addition, the study took women and historically marginalized population's experiences into account while investigating the STT in mathematics. Our engagement with the participants highlighted the importance of recognizing affective dimensions that contribute to the development of learner identity. As well, we found that the role of perceived competence and views on the nature of mathematics (i.e. our reconceptualization of 'vision of mathematics') play a role in that development. Furthermore, our exploratory study raised concern for the need of more well-developed informal and formal institutional mechanisms to support students' (perception of) success and development of learner identity within a mathematical community.

The study also revealed two findings that were notably different for PMs and SMTs. First, participants' consideration of grades as their conceptualization of success, or, with respect to a dimension of interest in this study, perceived competence. For example, among PM participants, there was not a consistent, direct relationship between perceived competence and grades received or previous mathematics accomplishment; i.e. higher grades or strong mathematical background did not always align with a PM participant's higher perceived competence. However, when SMT participants earned good grades, they often expressed equally high self-perception of competence that corresponded to their good grades. Second, when describing their experience with more abstract mathematics, SMT participants tended to criticize aspects of instruction (e.g. Evelyn's statement that the 'professor did little explaining on proofs'). Alternatively, PM participants were more apt to turn the focus on their role in the learning (e.g. Amelia's description of working with proofs and her desire to avoid memorization). The distinction between SMTs' perception on mathematics in secondary school compared with university calls attention to the need to investigate the way mathematics is taught in secondary schools (Geisler & Rolka, 2021).

To our knowledge, previous research has not emphasized the experiences of students in our population of interest, which included PM majors and double majors in PM and SMT and who are women or those from other marginalized populations, while they navigate the STT in mathematics. A more thorough understanding of this phenomenon—especially with regard to historically marginalized students in the mathematics major in the USA—holds promise for supporting not only students during the STT but also

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the departments and institutional mechanisms that facilitate their undergraduate mathematics programs. Moreover, investigating and addressing affective dimensions has the potential to help tell the stories of students who are historically marginalized in male-dominated mathematical spaces. Emotions as readout perspective (Buck, 1999) inform us that emotional responses can reveal whether 'all is well' (Reeve, 2008). How individuals interpret their own emotional states and construct meanings is context-dependent and ultimately affects their personal well-being (Reeve, 2008). Therefore, the three dimensions of the TMA (Di Martino & Zan, 2010), emotional disposition toward mathematics, perceived competence in mathematics and vision of mathematics, guide us to determine the potential needs of students by using their 'emotional readout' and to highlight supportive mechanisms they identify or deem necessary to navigate the mathematical spaces in which they participate.

Our focus on participants from one institution has potential to shed new light on the STT as it manifests in contexts and institutions in the USA, particularly because research on the STT in other countries does not equate well with similar investigations in the USA. For example, in a study of undergraduate mathematics majors in the STT in Germany, researchers found that the transition was immediately and emotionally experienced as students began their undergraduate studies in mathematics (Witzke *et al.,* 2018). Undergraduate mathematics programs in the USA typically entail a substantial general education/liberal arts component during the first 2 years of a 4-year program. Thus, students may not be initially confronted with the stark gap that exists between secondary school mathematics (often characterized by its empirically oriented nature) and university-level mathematics (which is more formal and abstract) that their counterparts in other countries encounter.

Based on this understanding of the nature of mathematics, we anticipated for undergraduate mathematics majors in the USA that the STT happens over time during their program of study as compared with tertiary students in other contexts. In our exploratory study, we found that first-year mathematics majors did not report an explicit difference between mathematics at school and at university. However, when we explicitly addressed features of the STT with participants, they revealed experiences steeped in their lack of confidence and limited agency to actively participate within their community of practice (Wenger, 1998). In the last decade, there has been a focus on attributes of women's success in postsecondary mathematics, despite the fact women have begun to outperform men in school mathematics in countries such as the UK (Solomon, 2007). Therefore, researchers must pursue understanding women's self-positioning as 'not belonging' despite their early success in mathematics (Solomon, 2007).

We recognize that although our qualitative research study investigated the experiences of approximately 20% of the population of interest at one large research university in the southeastern USA, there are limitations to the research. In particular, the two seminars entailed a relatively short amount of time (with regard to students' participation), and more time would be needed to capture more nuanced changes in their vision of mathematics, perceived competence and emotional dimensions that surface after engaging with participants during different time points in the STT. As well, we were not able to collect a full data profile for each participant. For example, students may not have been able to attend each seminar session (and therefore, we were unable to collect reflection diaries for sessions that were missed during the first seminar). This was further exacerbated by two intervening circumstances: the first seminar took place during a summer semester (and not all undergraduate majors take courses in summer) and the second took place during the start of the global COVID-19 pandemic. In addition, inclusion of more men and students from diverse backgrounds in the study would also provide additional insights.

Our future work on affective dimensions of students' experience in the STT entails a focus on efforts to facilitate women's development of their views on the nature of mathematics and their perceived competence regarding participation in mathematics, as well as proposing and investigating institutional mechanisms to support them as they navigate these efforts. In particular, investigating students' sense

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of belonging from a sociocultural standpoint to address mathematical competency and historically marginalized students' experiences will enable us to more deeply examine (and characterize) the affective dimensions that influence them. Furthermore, we believe that it is important to focus on participation and non-participation of learner identities within mathematical communities in which they are a member (Wenger, 1998).

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Conflicts of Interest

The authors declare that they have no conflicts of interest.

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APPENDIX

Sample Questions from the Interview Protocol.

1. Student information.

   a. What is your academic level (Fr, So, Jr, Sr)?
   b. What is your major?

2. Secondary school experience.

   a. How would you characterize (describe) the mathematics that you learned in middle and high school? (Witzke *et al.,* 2016; Witzke *et al.,* 2018)
   b. Did your secondary school grades influence your decision to major in mathematics? Why or why not? (Smith, 2012)

3. Decision to major in mathematics.

   a. Describe what helped you to make (or affected) the decision to major in mathematics. (Smith, 2012)

4. Current experience in the major.

   a. Tell me about your experience with the mathematics major thus far.
   b. Have you noted any change in what is required of your work in your mathematics courses (with respect to your cognitive/mathematical effort, not so much in actual assigned work) since beginning the mathematics major? (Thomas *et al.,* 2015)
   c. Please describe your engagement with fellow students as part of your academic work in the mathematics major. (Smith, 2012)
   d. What suggestions do you have about what your university could do to be supportive of women and underrepresented and/or marginalized students in the major? (Smith, 2012)
   e. How would you characterize your transition (the change) from school to university mathematics? (Witzke *et al.,* 2018)
   f. What suggestions do you have about what your institution could do to be supportive of women and underrepresented minorities in the major? (Smith, 2012)

5. Perceptions of mathematics.

Note: Provide each of the following on a separate page for the participant to read and reference during the interview.

Of the four descriptions provided, please select the one that best describes your belief about mathematics (or rank order the four) and tell me a bit about why this description resonates with you. (Grigutsch & Törner, 1998; Dubinsky & McDonald, 2001).

Application-aspect: This worldview describes the immediate relevance to application or the practical use of mathematics. Knowledge of mathematics is important to future life: mathematics either helps to solve everyday tasks and problems or it is useful to one's occupation. Apart from that, mathematics is of a general, fundamental use to society.

Formalism-aspect: This worldview of mathematics is characterized by strictness, exactness and precision on the terminological and language levels concerning thinking ('logical', 'objective' and 'flawless thinking'), argumentation and giving reasons and proof of statements (e.g. axiomatics and the strict deductive method).

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Process-aspect: In this worldview, mathematics is considered a process and an activity in thinking about problems and gaining knowledge. On the one hand, this cognitive process is about creating, inventing or reinventing (i.e. rediscovering) mathematics. On the other hand, it also includes the comprehension of facts and understanding connections. It requires thinking and arguing in regard to content as well as sudden and innovative ideas, intuition and experimenting. This [aspect] expresses the dynamic view of mathematics.

Schema-aspect: This worldview of mathematics is seen as a 'toolbox and bundle of formulas' and an idea oriented with algorithms and schemes. Mathematics is characterized by a collection of methods and rules that precisely determine how to solve a task. The consequence with dealing with mathematics is: doing mathematics consists of remembering and applying definitions, rules, formulas, facts and methods.

6. Perceived competence in mathematics.

   a. How confident do you feel about your mathematical ability at university? (Liston & O'Donoghue, 2010)
   b. Has your confidence in your mathematical ability changed since coming to university? (Liston & O'Donoghue, 2010)

7. Approaches to and assessment of learning mathematics

   a. What sort of instructional/teaching practices do your mathematics professors use in class? (And which of these are most beneficial/helpful to you?) (Di Martino & Gregorio, 2019)
   b. Describe how well you understand the mathematics course content you are currently studying in the major. [And please do so for each mathematics course you are taking.] (Liston & O'Donoghue, 2010; Thomas *et al.,* 2015)

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